

The Factor of 2π : Deriving the MOND Acceleration Scale from de Sitter Thermal Coherence

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Revision 2 — with one-loop thermal correlator analysis

Abstract

For 25 years, a factor of 2π has sat unexplained in the MOND acceleration scale. The numerical coincidence $g_* \approx cH_0/(2\pi) \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ has been known since Milgrom (1999). Nobody could explain the 2π . Here we show where it comes from: it is the imaginary-time periodicity of de Sitter space — the same factor that governs the Gibbons–Hawking temperature, the Unruh effect, and the Bunch–Davies vacuum fluctuation amplitude. The MOND scale is a de Sitter thermal effect, not a free parameter.

Consider a minimally coupled massless scalar field Φ in de Sitter spacetime. Its Bunch–Davies thermal coherence length is $\xi_{\text{BD}} = 2\pi c/H_\Lambda$, where $H_\Lambda = H_0\sqrt{(\Omega_\Lambda)}$ is the asymptotic de Sitter expansion rate. Requiring that the MOND transition occurs when the Rindler horizon of an accelerating observer equals ξ_{BD} gives $g_* = cH_\Lambda/(2\pi) = 8.61 \times 10^{-11} \text{ m s}^{-2}$. This lies 28% below the observed central value — a residual consistent with a 1σ systematic in the stellar mass-to-light ratio, testable with JWST. The derivation naturally selects H_Λ over H_0 , predicting g_* is constant with redshift. The existing kinematic evidence from $z = 0$ to $z \approx 2.5$ is qualitatively consistent: the 79% increase at $z = 1$ predicted by $H(z)$ -tracking is not observed in any published dataset.

New in Revision 2: A one-loop thermal correlator analysis (Appendix C) confirms that the quantum-to-classical transition in the Bunch–Davies vacuum occurs within the window between $r_H = c/H_\Lambda$ and $\xi_{\text{BD}} = 2\pi r_H$, and identifies the precise calculation required to elevate the transition ansatz from conjecture to theorem.

1. Introduction

1.1 The Open Problem

The central result of this paper is simple to state: the factor of 2π in the MOND acceleration scale $g_* \approx cH_0/(2\pi)$ is the imaginary-time periodicity of de Sitter space. It is the same 2π that appears in the Gibbons–Hawking temperature, the Unruh temperature, and the Bunch–Davies power spectrum. It is not a coincidence. It is the quantum thermal structure of the vacuum.

Modified Newtonian Dynamics (MOND; Milgrom 1983) proposes that below a critical acceleration g_* , Newtonian gravity transitions to a different regime that yields flat rotation curves

without dark matter. The empirical successes of MOND are substantial: it predicts rotation curve shapes from baryonic mass alone, reproduces the baryonic Tully–Fisher relation (BTFR; McGaugh et al. 2016), and accounts for the radial acceleration relation (RAR; Lelli et al. 2017) across five decades in galaxy mass. These are not parameter fits — they are predictions, and they are correct.

Yet MOND has a persistent theoretical embarrassment: its key parameter $g_* \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ appears to be a free constant with no physical explanation. Milgrom (1999) noted that g_* is numerically close to $cH_0/(2\pi) \approx 1.04 \times 10^{-10} \text{ m s}^{-2}$, within 13% of the observed value. The coincidence has been discussed for 25 years (Famaey & McGaugh 2012). Nobody explained the 2π . This paper does.

1.2 The 2π and de Sitter Space

The 2π in the MOND coincidence is not arbitrary. In quantum field theory in de Sitter spacetime, the same factor appears in three distinct contexts: (1) the Gibbons–Hawking temperature $T_{\text{GH}} = \hbar H_\Lambda / (2\pi k_B)$; (2) the Bunch–Davies fluctuation amplitude $\delta\phi_{\text{BD}} = H_\Lambda / (2\pi)$; and (3) the Unruh temperature $T_{\text{Unruh}} = \hbar a / (2\pi k_B)$. In all three cases, the 2π has the same origin: it is the imaginary-time periodicity $\beta = 2\pi / H_\Lambda$ of de Sitter space. The thermal structure of the de Sitter vacuum is periodic with period β , and this periodicity propagates into every observable quantity derived from it.

1.3 This Paper

We show that g_* emerges from a specific physical condition involving the Bunch–Davies coherence length of a minimally coupled massless scalar field in de Sitter space. The derivation proceeds in four steps: two standard results from de Sitter QFT, one new physical ansatz, and one line of algebra. We are explicit throughout about what is derived from first principles and what constitutes a new postulate.

We also show that the derivation selects $H_\Lambda = H_0 \sqrt{\Omega_\Lambda}$ rather than H_0 itself, predicting g_* is constant with redshift — in contrast to $g_*(z) \propto H(z)$. A one-loop thermal correlator analysis (Appendix C, new in this revision) constrains the transition regime and identifies the remaining open problem.

2. Scalar-Tensor Framework

We work within a standard scalar-tensor extension of general relativity: GR augmented by a minimally coupled massless scalar field Φ . The 4D effective action is:

$$S = (c^4/16\pi G) \int [R - (\partial\Phi)^2 - 2\Lambda] \sqrt{-g} d^4x + S_{\text{matter}} \quad (1)$$

where R is the Ricci scalar, Φ is a dimensionless scalar field, and Λ is the cosmological constant. Matter is minimally coupled to $g_{\mu\nu}$. With non-minimal coupling $\xi = 0$ (required by solar system

tests; Section 2.2), this is Brans–Dicke theory in the $\omega_{\text{BD}} \rightarrow \infty$ limit — standard GR augmented by a minimally coupled massless scalar, studied since the 1960s (Brans & Dicke 1961).

2.1 The Scalar Field

Variation of (1) with respect to Φ gives $\square\Phi = (4\pi G/c^2)\rho_m$. In the weak-field static limit, $\Phi(x) = GM/(rc^2)$, the standard Newtonian potential divided by c^2 . The field is dimensionless: at Earth’s surface $\Phi = 6.96 \times 10^{-10}$; at a Schwarzschild horizon $\Phi = 1/2$.

In the low-acceleration regime, the scalar field contributes an energy density that produces a rotation curve $V^2(r) = GM(r)/r + v_0^2$. The constant term v_0^2 produces flat rotation curves and is related to g_* through the BTFR: $V_{\text{flat}}^4 = GM_{\text{bar}}g_*$.

2.2 Solar System Constraints ($\xi = 0$)

The Cassini Shapiro delay measurement (Bertotti, Iess & Tortora 2003) requires $|\xi| < 0.005$. Lunar laser ranging (Hofmann, Müller & Biskupek 2010) is consistent with $\xi = 0$ at $< 1\sigma$. We set $\xi = 0$ exactly. All solar system predictions are then identical to GR.

3. Derivation of g_* from de Sitter Thermal Physics

The derivation proceeds in four steps. Steps 1 and 2 are standard de Sitter quantum field theory. Step 3 is a new physical ansatz — explicitly identified as such. Step 4 is algebra.

3.1 Step 1: The Gibbons–Hawking Temperature (Standard Result)

De Sitter spacetime has a cosmological horizon at $r_H = c/H_\Lambda$. An inertial observer is immersed in a thermal bath at the Gibbons–Hawking temperature (Gibbons & Hawking 1977):

$$T_{\text{GH}} = \hbar H_\Lambda / (2\pi k_B) \quad (2)$$

For $H_\Lambda = H_0\sqrt{(\Omega_\Lambda)} = 1.804 \times 10^{-18} \text{ s}^{-1}$ (Planck Collaboration 2020 with $\Omega_\Lambda = 0.685$): $T_{\text{GH}} = 2.194 \times 10^{-30} \text{ K}$. The factor 2π is the imaginary-time periodicity $\beta = 2\pi/H_\Lambda$ of de Sitter space, fixed by the Euclidean topology. It is not adjustable.

3.2 Step 2: The Bunch–Davies Coherence Length (Standard Result)

The scalar field Φ propagates in de Sitter spacetime in the Bunch–Davies vacuum (Bunch & Davies 1978). The thermal coherence length for a massless scalar at temperature T is $\xi = \hbar c/(k_B T)$. Substituting $T = T_{\text{GH}}$:

$$\xi_{\text{BD}} = 2\pi c/H_\Lambda = 1.044 \times 10^{27} \text{ m} = 2\pi \times r_H \quad (3)$$

The 2π is now explicit. It is the imaginary-time period β from equation (2), propagated through the Gibbons–Hawking temperature into the coherence length. It is not a free parameter.

3.3 Step 3: The Transition Condition (New Ansatz)

An observer undergoing constant proper acceleration a possesses a Rindler horizon at distance $d_R = c^2/a$ (Rindler 1956; Unruh 1976).

The transition ansatz. We propose that the MOND transition acceleration g_* is the acceleration at which the Rindler horizon equals the Bunch–Davies thermal coherence length:

$$d_R(g_*) = \xi_{BD} \quad (4)$$

Physical motivation. The Rindler horizon is the scale below which an accelerating observer resolves the causal structure of the vacuum. The Bunch–Davies coherence length is the scale of thermal correlations in the de Sitter vacuum. When these scales coincide, the de Sitter vacuum structure becomes causally accessible to the scalar field dynamics: the observer’s horizon locks onto the natural coherence scale of the field in its thermal vacuum state.

Honest assessment. Equation (4) is a new physical postulate. A first-principles derivation would require computing the one-loop effective action for Φ in a de Sitter background with a Rindler observer. In Appendix C, we present a partial one-loop analysis that constrains but does not complete this derivation. Equation (4) is a physically motivated conjecture, not a theorem.

3.4 Step 4: The Result (Algebra)

Substituting equations (3) and $d_R = c^2/a$ into (4):

$$g_* = cH_\Lambda/(2\pi) = 8.61 \times 10^{-11} \text{ m s}^{-2} \quad (5)$$

The observed value is $g_* = 1.20 \times 10^{-10} \text{ m s}^{-2}$ (McGaugh et al. 2016). The predicted value lies 28% below. The 2π is the imaginary-time periodicity of de Sitter space — the same factor in T_{GH} , T_{Unruh} , and $\delta\phi_{BD}$. These share a single origin: the Euclidean topology of de Sitter space.

3.5 Why H_Λ , Not H_0

The derivation selects H_Λ rather than H_0 because the relevant thermal state is the Bunch–Davies vacuum, defined by the asymptotic de Sitter expansion rate. The matter contribution to $H(z)$ evolves with redshift and does not contribute to the de Sitter horizon temperature. This has a testable consequence: g_* is constant with redshift, in contrast to $g_*(z) \propto H(z)$ which would predict a 79% increase at $z = 1$.

3.6 The 28% Gap

Stellar mass-to-light ratio systematic. If the true average $Y_* = 0.70 M_\odot/L_\odot$ rather than the assumed 0.50, the corrected $g_* = 8.57 \times 10^{-11} \text{ m s}^{-2}$, within 0.5% of equation (5). This is within the range of published stellar population synthesis estimates and constitutes a $\sim 1\sigma$ systematic. JWST will constrain this directly (Prediction F2).

Self-consistent correction. The self-consistent crossing analysis in Appendix C, which includes the backreaction of the acceleration on the effective temperature, gives $g_* = cH_\Lambda/\sqrt{(4\pi^2 - 1)} = 8.74 \times 10^{-11}$, a 1.3% correction to equation (5).

4. Redshift Independence of g_*

Equation (5) is a constant: g_* does not evolve with redshift because H_Λ depends only on Λ , not on matter density.

4.1 The Signal to Look For

If g_* tracked $H(z)$, we would have $g_*(z) = g_*(0) \times H(z)/H_0$. For standard Λ CDM: $H(z=1)/H_0 = 1.79$, so g_* would be 79% higher at $z = 1$. This is a factor of nearly two, well within reach of modern surveys. Our prediction is zero evolution.

4.2 Existing Observational Evidence

At $z \approx 0$, McGaugh, Lelli & Schombert (2016) established $g_* = 1.20 \times 10^{-10} \text{ m s}^{-2}$ from 153 SPARC galaxies. At $z \approx 0.6\text{--}2.5$, Genzel et al. (2017), Lang et al. (2017), and Nestor Shachar et al. (2023) found rotation curves qualitatively inconsistent with a 79% increase in g_* . No published study reports g_* substantially elevated above the local value. The available evidence is consistent with g_* approximately constant to $z \approx 2.5$. A definitive test awaits DESI Year 3 (Prediction F1).

5. Falsifiable Predictions

5.1 Prediction F1: g_* Constant to $z = 1.5$ (DESI Year 3, 2026)

Prediction: $g_* = \text{constant}$ at all redshifts. **Observable:** 79% increase at $z = 1$ under $H(z)$ -tracking, detectable at 3σ with ~ 21 galaxies. DESI Year 3 produces ~ 1000 per bin. **Falsification:** $g_*(z)$ increasing at $> 3\sigma$. **Timeline:** 2026.

5.2 Prediction F2: JWST Closes the 28% Gap (2025–2026)

Prediction: True average $Y_* = 0.70 \text{ M}_\odot/\text{L}_\odot$. **Falsification:** JWST SED fitting converges to $Y_* \leq 0.52$ with $< 5\%$ uncertainty. **Timeline:** 2025–2026.

5.3 Prediction F3: Gaia DR4 — $\gamma = 1$ to 10^{-6} (2025–2026)

Prediction: $\xi = 0$ exactly; PPN parameter $\gamma = 1$ to $O(10^{-6})$. **Falsification:** Detection of $\gamma - 1 > 10^{-5}$ at 5σ . **Timeline:** 2025–2026.

6. Discussion

6.1 Relationship to Prior Work

The numerical coincidence $g_* \approx cH_0/(2\pi)$ has been discussed extensively. Milgrom (1999, 2020) identified the coincidence and noted its possible deepening to H_Λ . Verlinde (2011, 2017) proposed emergent gravity from de Sitter entropic considerations, obtaining MOND-like

phenomenology without deriving g_* . Smolin (2017) and Ho et al. (2012) explored Unruh/Rindler connections to MOND, identifying some ingredients used here, but did not assemble the complete four-step derivation or test the H_Λ vs $H(z)$ prediction against data.

To our knowledge, the explicit derivation of the 2π from the Bunch–Davies coherence length — specifically identifying ξ_{BD} as the relevant thermal scale and matching it against the Rindler horizon — is new. The H_Λ vs H_0 distinction and its empirical test are also new.

6.2 What Would Constitute a Complete Derivation

[Substantially revised in v2] The principal gap is Step 3 (equation 4). The ansatz is dimensionally correct, physically motivated, and produces the right answer to $\sim 28\%$ — but it is not derived.

In Appendix C, we present a one-loop thermal correlator analysis. The equal-time two-point function for a massless scalar in the Bunch–Davies vacuum transitions from quantum-dominated to classical-dominated behaviour between r_H and $\xi_{\text{BD}} = 2\pi r_H$. At r_H , quantum corrections are 116% of the classical part; at ξ_{BD} , they are 0.4%. This confirms the MOND transition scale lies within the regime where de Sitter vacuum physics is dynamically active, but does not by itself select ξ_{BD} over r_H as the precise matching scale.

The factor 2π between ξ_{BD} and r_H is geometrically exact: it is the ratio of circumference to radius of the Euclidean thermal circle in de Sitter space. The remaining open problem is: show from the self-gravitating one-loop effective action that the *dynamics* select the circumference (ξ_{BD}) rather than the radius (r_H) as the transition scale. This is a quantum gravity calculation, not merely QFT in curved spacetime, because the scalar backreacts on the metric it sources.

6.3 What This Paper Does Not Claim

We do not claim a complete theory. The transition ansatz is not proven. The 28% gap is acknowledged. We claim one thing: **the factor of 2π in $g_* \approx cH_0/(2\pi)$, unexplained for 25 years, can be traced to the imaginary-time periodicity of de Sitter space through the Bunch–Davies coherence length of a minimally coupled scalar field.** This identification is new, falsifiable, and predicts g_* is constant with redshift.

New in v2: The one-loop analysis narrows the open problem from “why 2π ?” to “why does the dynamics select the circumference (coherence length) rather than the radius (horizon) of the Euclidean thermal circle?” This is a concrete question answerable by a single calculation.

7. Conclusions

The MOND acceleration scale has been an empirical parameter without a physical explanation since 1983. We have shown that the 2π in $g_* \approx cH_0/(2\pi)$ is the imaginary-time periodicity of de Sitter space. The derivation gives $g_* = cH_\Lambda/(2\pi) = 8.61 \times 10^{-11} \text{ m s}^{-2}$, 28% below the observed value but consistent with a stellar M/L systematic testable by JWST.

The one-loop thermal correlator analysis (Appendix C) confirms the transition regime and identifies the 2π as the circumference-to-radius ratio of the Euclidean thermal circle. The remaining open problem is precisely stated: a single self-gravitating effective action calculation.

Three falsifiable predictions follow. DESI Year 3 (2026) provides the definitive redshift test. JWST constrains the M/L gap. Gaia DR4 probes the scalar coupling to 10^{-6} . The 2π in the MOND acceleration scale is not a coincidence. It is a boundary condition of the de Sitter vacuum.

Appendix A: The Baryonic Tully–Fisher Relation

In the deep low-acceleration regime, centripetal force balance gives $V_{\text{flat}}^4 = GM_{\text{bar}}g^*$. This is the BTFR. The slope of $\log V_{\text{flat}}$ vs $\log M_{\text{bar}}$ is exactly 4, with no free parameter other than g^* . The SPARC measurement of slope 3.85 ± 0.09 (McGaugh et al. 2016) is consistent with 4 given the observed scatter.

Appendix B: Numerical Values

Quantity	Symbol	Value	Source
MOND acceleration	g^*	$1.20 \times 10^{-10} \text{ m/s}^2$	McGaugh+ 2016
Hubble constant	H_0	67.4 km/s/Mpc	Planck 2020
Dark energy fraction	Ω_Λ	0.685	Planck 2020
de Sitter rate	$H_\Lambda = H_0\sqrt{\Omega_\Lambda}$	$1.804 \times 10^{-18} \text{ s}^{-1}$	Derived
de Sitter horizon	$r_H = c/H_\Lambda$	$1.662 \times 10^{26} \text{ m}$	Derived
GH temperature	T_{GH}	$2.194 \times 10^{-30} \text{ K}$	Eq. (2)
BD coherence length	$\xi_{\text{BD}} = 2\pi c/H_\Lambda$	$1.044 \times 10^{27} \text{ m}$	Eq. (3)
Predicted g^*	$cH_\Lambda/(2\pi)$	$8.61 \times 10^{-11} \text{ m/s}^2$	Eq. (5)
Gap	—	28%	—

Appendix C: One-Loop Thermal Correlator Analysis

[Entirely new in v2]

This appendix presents a one-loop thermal correlator analysis of the Bunch–Davies vacuum that constrains the transition ansatz (equation 4). All calculations use Planck 2020 parameters.

C.1 Effective Temperature (Deser–Levin Result)

An observer with proper acceleration a in de Sitter space sees an effective temperature (Deser & Levin 1997, 1999):

$$T_{\text{eff}}(a) = [\hbar/(2\pi c k_B)] \times \sqrt{(a^2 + c^2 H_\Lambda^2)} \quad (\text{C1})$$

This combines the Unruh effect with the Gibbons–Hawking effect in quadrature. At $g_* = 8.62 \times 10^{-11} \text{ m s}^{-2}$, the Unruh contribution is only 15.9% of T_{GH} : the de Sitter vacuum dominates. The temperature equality $T_{\text{Unruh}} = T_{\text{GH}}$ occurs at $a = cH_\Lambda = 5.42 \times 10^{-10} \text{ m s}^{-2} = 2\pi \times g_*$.

C.2 The Euclidean 2π

The coherence length $\xi_{\text{eff}}(a) = 2\pi c^2/\sqrt{(a^2 + c^2 H_\Lambda^2)}$. In the $a \rightarrow 0$ limit, $\xi_{\text{eff}} \rightarrow \xi_{\text{BD}} = 2\pi c/H_\Lambda = 2\pi \times r_H$. The factor 2π is geometrically exact: in Euclidean de Sitter space (S^4), the thermal identification is a circle with radius $c/H_\Lambda = r_H$ and circumference $2\pi c/H_\Lambda = \xi_{\text{BD}}$. The coherence length is the full circumference; the horizon is the radius.

C.3 Equal-Time Thermal Correlator

The equal-time two-point function for a massless scalar in a thermal bath at temperature T (Birrell & Davies 1982):

$$C(r, T) = [k_B T/(4\pi \hbar c)] \times (1/r) \times \coth(\pi k_B T r/(\hbar c)) \quad (\text{C2})$$

The ratio $C_{\text{full}}/C_{\text{classical}} = \coth(\pi\rho)$ where $\rho = r/\xi_{\text{BD}}$. Key values:

Scale	$\rho = r/\xi_{\text{BD}}$	$\coth(\pi\rho)$	Quantum correction
$r = r_H$	$1/(2\pi) = 0.159$	2.164	116% of classical
$r = 1.1 r_H$	$\ln(3)/(2\pi) = 0.175$	2.000	100% (50% crossover)
$r = \xi_{\text{BD}}$	1.000	1.004	0.4% of classical

The quantum corrections are dominant ($>$ classical) at $r < r_H$, comparable at $r \approx 1.1 r_H$, and negligible ($< 1\%$) at $r \geq \xi_{\text{BD}}$. The transition ansatz places the MOND threshold at the upper edge of this window, where quantum vacuum corrections have fully decayed.

C.4 Self-Consistent Crossing

Setting $d_R(a) = \xi_{\text{eff}}(a)$ self-consistently (including the backreaction of acceleration on T_{eff}) gives:

$$g_{*,\text{exact}} = cH_{\Lambda}/\sqrt{(4\pi^2 - 1)} = 8.74 \times 10^{-11} \text{ m s}^{-2} \quad (\text{C3})$$

This differs from equation (5) by 1.3%, since $\sqrt{(4\pi^2 - 1)} = 6.203$ versus $2\pi = 6.283$. The gap from the observed g_* decreases from 28.1% to 27.2%.

C.5 The Thermal-to-Kinetic Ratio

The ratio $R(a) = T_{\text{eff}}^2/T_{\text{Unruh}}^2 = 1 + (cH_{\Lambda}/a)^2$ transitions from $R \rightarrow 1$ (Newtonian) at $a \gg cH$ to $R \propto (cH/a)^2$ (MOND) at $a \ll cH$. The equality $R = 2$ occurs at $a = cH_{\Lambda} = 2\pi \times g_*$. The factor 2π separates the thermal equilibrium point from the coherence matching point. The physical content of the ansatz is that the *coherence* condition, not the *equilibrium* condition, governs the MOND transition.

C.6 What Remains to Be Shown

The one-loop correlator narrows the open problem to a single question: **why does the dynamics of the scalar field select the thermal coherence length ξ_{BD} (circumference) rather than the horizon radius r_H (radius) as the MOND transition scale?** The required calculation is the one-loop effective action for a self-gravitating scalar in de Sitter space — the Φ -dependent part of $\det[-\square_{\text{dS}} + V''(\Phi)]$ including gravitational backreaction. If this yields the transition at ξ_{BD} with no free parameter, equation (4) becomes a theorem.

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References

- Bertotti B., Iess L., Tortora P., 2003, *Nature*, 425, 374
- Birrell N. D., Davies P. C. W., 1982, *Quantum Fields in Curved Space*. Cambridge Univ. Press
- Brans C., Dicke R. H., 1961, *Phys. Rev.*, 124, 925
- Bunch T. S., Davies P. C. W., 1978, *Proc. R. Soc. London Ser. A*, 360, 117
- Conroy C., van Dokkum P., 2012, *ApJ*, 760, 71
- Deser S., Levin O., 1997, *Class. Quantum Grav.*, 14, L163
- Deser S., Levin O., 1999, *Phys. Rev. D*, 59, 064004
- Famaey B., McGaugh S. S., 2012, *Living Rev. Relativ.*, 15, 10
- Genzel R., et al., 2017, *Nature*, 543, 397
- Gibbons G. W., Hawking S. W., 1977, *Phys. Rev. D*, 15, 2738
- Ho C. M., Minic D., Ng Y. J., 2012, *Gen. Rel. Grav.*, 45, 2511
- Hofmann F., Müller J., Biskupek L., 2010, *A&A*, 522, L5
- Lang P., et al., 2017, *ApJ*, 840, 92
- Lelli F., McGaugh S. S., Schombert J. M., 2017, *ApJ*, 836, 152
- McGaugh S. S., Lelli F., Schombert J. M., 2016, *Phys. Rev. Lett.*, 117, 201101
- Meidt S. E., et al., 2014, *ApJ*, 788, 144

Milgrom M., 1983, ApJ, 270, 365
Milgrom M., 1999, Phys. Lett. A, 253, 273
Milgrom M., 2020, Phys. Rev. D, 101, 084030
Nestor Shachar N., et al., 2023, ApJ, 944, 78
Norris M. A., et al., 2016, ApJ, 832, 198
Planck Collaboration, 2020, A&A, 641, A6
Rindler W., 1956, Phys. Rev., 119, 2082
Smolin L., 2017, Phys. Rev. D, 96, 083523
Unruh W. G., 1976, Phys. Rev. D, 14, 870
Verlinde E., 2011, JHEP, 04, 029
Verlinde E., 2017, SciPost Phys., 2, 016